Q1.

(a) In an iron rod the particles vibrate. If one end of an iron rod is heated, the vibrating particles transfer energy to neighbouring particles which are **not** vibrating so violently. What is this process called?

……

1 mark

(b) An electric immersion heater is put at the bottom of a large tank of water.

The water next to the heater becomes warm.

(i) What will happen to the warmed water next to the heater?
Give a reason for your answer.

……

……

……

2 marks

(ii) Why can heat **not** be transferred in this way in an iron rod?

……

……

1 mark

(c) In a liquid, some of the particles have enough kinetic energy to escape from the surface. This process happens even when the liquid is well below its boiling point.

(i) What is this process called? ……

1 mark

(ii) How will this affect the temperature of the liquid left in the container?

……

1 mark

Maximum 6 marks

Q1.

(a) conduction

1 (L7)

(b) (i) it rises

*accept 'it forms a convection current' **or** 'it floats to the top'*

1 (L7)

any **one** from

- it expands
  accept 'the molecules move further apart'
- it becomes less dense
  accept 'the particles move more quickly'

1 (L7)

(ii) any **one** from

- the atoms or particles in a solid cannot move
  *accept 'the atoms cannot move around **or** are fixed'*
- the atoms or particles are bonded tightly
  accept 'particles are bonded'
- iron is not fluid
  accept 'iron has a fixed shape'
  *do **not** accept 'iron is a solid'*

1 (L7)

(c) (i) evaporation

accept 'evaporating'

1 (L6)

(ii) any **one** from

- it gets colder
- it decreases
  accept 'it loses heat'

1 (L7)

[6]